DEFINITION and CORE FUNCTION

1.A **read-only industrial edge gateway** that sits inside a data center network and **collects power and cooling telemetry**, validates it locally, and securely forwards it to Enervia's cloud optimization engine.

- It is **not a controller**
- It is **not in the safety loop**
- It is **not required for uptime**

2. What it connects to

The edge device connects only to **existing systems**, using **standard, non-invasive interfaces**.

Physical / Network Connections

Connection	Connects To	Purpose
Ethernet (LAN)	Data center network	Primary data access
Ethernet (SNMP)	UPS systems	Load, battery status
Ethernet (BACnet / Modbus TCP)	BMS / chillers	Cooling & thermal data
RS-485 (Modbus RTU)	Power meters	Facility & chiller power
Analog IN (4–20 mA / 0–10 V)	Temp / airflow sensors (if needed)	Rack & CRAC sensing

No proprietary buses.
No vendor lock-in.

3. What signals it reads

The device reads **only the minimum signals required for real optimization**.

Signals Handled

Signal	Source	Why Enervia Needs It
Rack inlet temperature	Sensor / BMS	Thermal safety boundary
Rack outlet temperature	Sensor / BMS	Cooling effectiveness
Airflow proxy (fan speed / ΔP)	CRAC / CRAH	Airflow imbalance detection
Chiller power (kW)	Power meter / BMS	Largest energy cost lever
Total facility power (kW)	Main meter	Peak demand & savings
UPS load (kW / %)	UPS	Stability & peak buffering
Time + tariff	Config / cloud	Cost-aware decisions

All signals are:

- read-only
- timestamped
- range-validated at the edge

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4. How it stays safe in a live data center

This is the **most important part**.

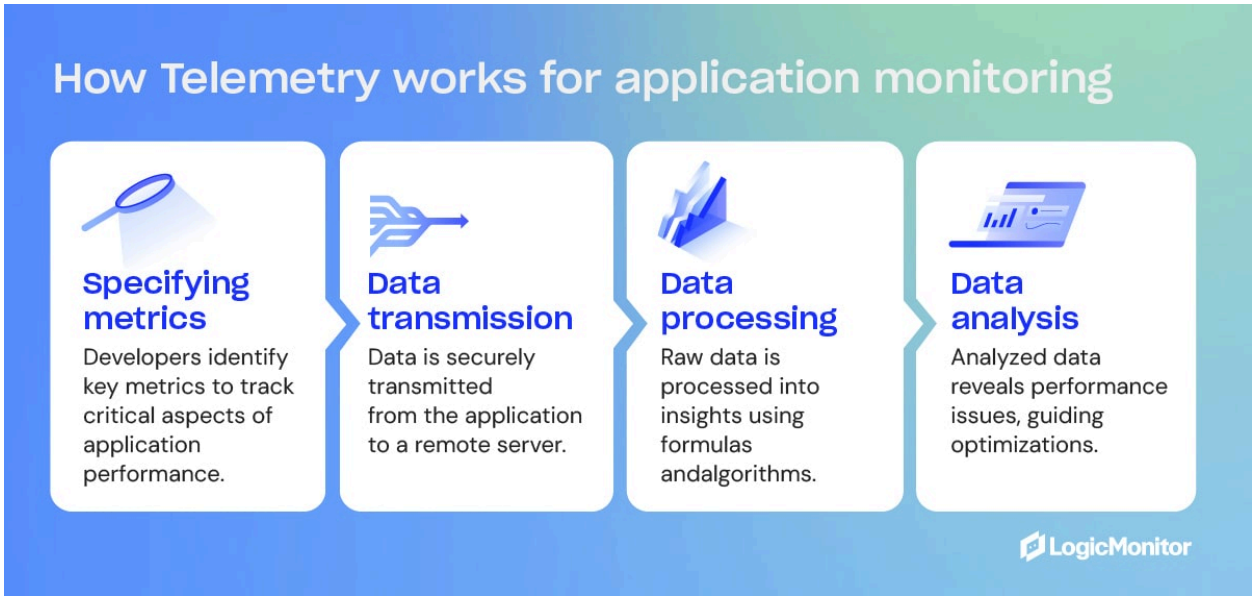
Safety & Isolation Principles

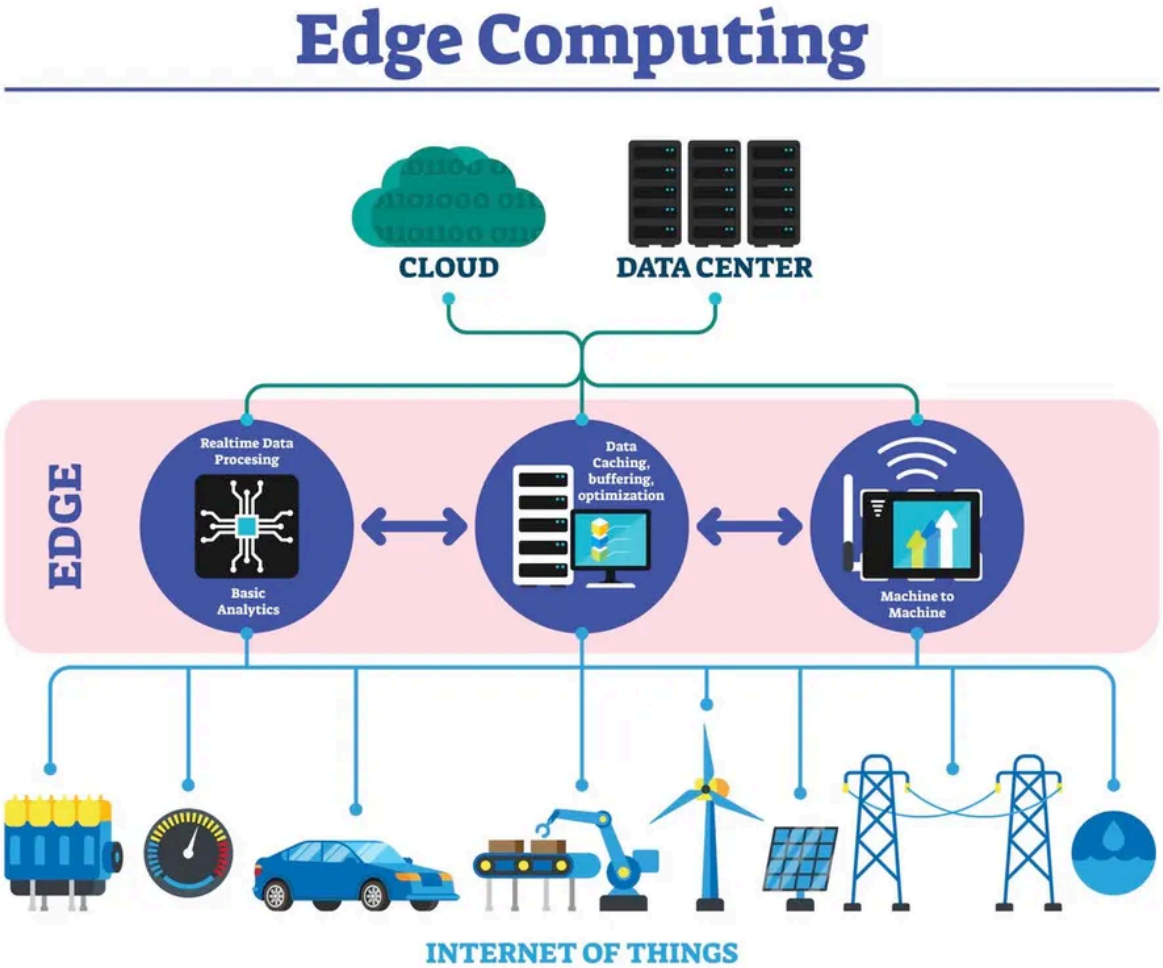
1. **Read-only by design**
    - No digital outputs
    - No write commands
    - No control loops
2. **Out of the safety chain**

    - Chillers, UPS, protection logic run independently
    - Enervia is never required for operation
3. **Fail-safe behavior**
    - If device fails → nothing changes in the data center
    - If data is missing → Enervia recommends nothing
4. **Network isolation**
    - Separate VLAN / firewall rules
    - Cannot issue control packets even if compromised
5. **Advisory-only (Phase-1)**
    - Outputs recommendations, not actions
    - Human approval always required

ONE LINE DEFINITION

The Enervia Edge Gateway is a read-only industrial device that safely extracts power and cooling data from a data center and feeds it to Enervia's AI engine, without touching any control or safety systems.


How Telemetry works for application monitoring
Specifying metrics
Developers identify key metrics to track critical aspects of application performance.
Data transmission
Data is securely transmitted from the application to a remote server.
Data processing
Raw data is processed into insights using formulas andalgorithms.
Data analysis
Analyzed data reveals performance issues, guiding optimizations.
LogicMonitor


Edge Computing
CLOUD
DATA CENTER
EDGE
Realtime Data Procesing
Basic Analytics
Data Caching, buffering, optimization
Machine to Machine
INTERNET OF THINGS

Technical Definition and features

Phase-1 Objective

Build a **safe, read-only energy optimization system** for **data & AI data centers** that reduces **cooling energy and peak electricity cost** using deterministic logic and AI-ready pipelines—**without controlling any equipment**.

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1. System Scope (Hard Boundary)

Included

- Cooling energy optimization (analysis + recommendations)
- Peak-tariff cost optimization
- Rack-level thermal safety checks
- Power visibility (facility, cooling, UPS)
- Advisory outputs only

Excluded

- Direct control of chillers, CRACs, UPS, breakers
- Replacement of DCIM/BMS
- Any safety-critical control loop
- Autonomous actions

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2. Phase-1 Architecture (Technical)

A. Edge Layer (On-site)

Component: Enervia Edge Gateway (EEG-01)

Responsibilities

- Read data from existing systems
- Validate ranges & timestamps
- Buffer and forward data
- Never issue control commands

Interfaces

- Ethernet (BACnet IP / Modbus TCP) → BMS, chillers
- RS-485 (Modbus RTU) → power meters
- Ethernet (SNMP) → UPS
- Optional analog in (4–20 mA / 0–10 V) → temp/airflow sensors

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B. Cloud / Compute Layer

Component: Enervia Optimization Engine (EOE-01)

Responsibilities

- Time-series ingestion
- Rule-based optimization
- Cost-aware decision logic
- Savings estimation
- Explainable recommendations

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3. Signals (Minimum Required Set)

Category	Signal	Unit
Rack	Inlet temperature	°C
Rack	Outlet temperature	°C
Air	Fan speed / airflow proxy	% / RPM
Cooling	Chiller power	kW
Power	Total facility power	kW
Power	UPS load	kW / %
Context	Time, tariff	hh:mm, ₹/kWh

All signals are read-only.

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4. Core Algorithms (Phase-1)

A. Thermal Safety Guard

- Hard constraints on rack inlet temperature
- ΔT trend monitoring (block actions on rapid rise)
- Global gatekeeper for all recommendations

B. Overcooling Detection

- Identifies excess cooling (low supply temps, low ΔT)
- Flags candidates for safe reduction

C. Peak-Tariff Detection

- Identifies high-cost windows

- Prioritizes actions only during economic peaks

D. Cooling Reduction Feasibility

- Allows reduction only if:
    - All racks thermally safe
    - No abnormal temperature trends
    - System within stability margins

E. Savings Estimation

- Computes kW reduction
- Computes ₹ savings per interval
- Aggregates daily/weekly impact

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5. Decision Loop (Deterministic)

1. **Observe:** ingest signals
2. **Validate:** range + freshness checks
3. **Assess:** thermal safety + cost window
4. **Decide:** allow/block optimization
5. **Recommend:** action, window, expected savings
6. **Verify:** no safety violation flags

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6. Outputs (Phase-1)

Recommendation Object

- Action: reduce cooling load (percentage)
- Time window
- Expected kW saved
- Expected ₹ saved
- Safety status (SAFE / BLOCKED)
- Explanation (which rules triggered)

No commands are executed.

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7. Safety Model (Non-Negotiable)

- Read-only hardware and software
- No dependency for uptime
- No override of native controls
- Fail-silent behavior
- Human approval required

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8. Success Criteria

Phase-1 is successful if it demonstrates:

- Repeated peak-hour cost reduction opportunities
- Zero thermal safety violations
- Explainable, deterministic decisions

- Verifiable kW and ₹ savings from real data

SOFTWARE MAPPING

1)High-Level Software Architecture

```
Edge Ingestion
    ↓
Data Validation
    ↓
Time-Series Store
    ↓
Safety Engine (Global Gatekeeper)
    ↓
Optimization Engines
    ↓
Savings Engine
    ↓
Recommendation Generator
```

MODULE 1: Data Ingestion Service

Module Name

`enervia_ingestion`

What it does

- Receives data from edge gateway
- Parses timestamps

- Normalizes units
- Pushes data to internal pipeline

Inputs

- JSON / CSV streams
- Signals:
    - rack temps
    - airflow proxy
    - chiller power
    - facility power
    - UPS load
    - tariff

Outputs

- Clean, structured time-series records

Why this module exists

All downstream logic depends on **clean, consistent data**.
This must be isolated.

MODULE 2: Data Validation & Sanity Checker

Module Name

`enervia_validation`

What it does

- Range checks
- Missing data detection
- Stale data detection
- Spike detection

Key Rules

- Rack inlet > 45°C → INVALID
- Negative power → INVALID
- Sudden jump > threshold → FLAG

Outputs

- Validated data
- Validation flags

Why this module exists

No optimization is allowed unless this module approves the data.

MODULE 3: Time-Series State Manager

Module Name

`enervia_state_store`

What it does

- Maintains rolling windows (15 min, 1 hr, 24 hr)
- Computes trends

- Stores last known safe state

Derived Metrics

- ΔT trends
- Rate of temperature change
- Moving averages

Why this module exists

Optimization decisions require **history**, not just instant values.

MODULE 4: Thermal Safety Engine (GLOBAL GATEKEEPER)

Module Name

`enervia_safety_engine`

What it does

- Enforces all hard constraints
- Has veto power over every optimization

Checks

- Rack inlet temp < limit
- ΔT not rising rapidly
- No unresolved validation flags

Outputs

- `SAFE` / `BLOCKED`
- Reason codes

Why this module exists

This is **non-negotiable IP**.
Every other module depends on this.

MODULE 5: Tariff & Time Context Engine

Module Name

`enervia_cost_context`

What it does

- Classifies time as:
  - PEAK
  - NORMAL
  - OFF-PEAK
- Computes cost severity index

Inputs

- Time
- Tariff (₹/kWh)

Outputs

- Cost context object

Why this module exists

Energy ≠ cost.
This module injects **economic intelligence**.

MODULE 6: Overcooling Detection Engine

Module Name

`enervia_overcooling_detector`

What it does

- Detects excess cooling
- Flags inefficient operating points

Inputs

- Rack temps
- Supply/return temps
- ΔT

Outputs

- Overcooling score
- Reduction feasibility hint

Why this module exists

This explains **where waste exists**, not just that it exists.

MODULE 7: Cooling Optimization Engine

Module Name

`enervia_cooling_optimizer`

What it does

- Decides *if* cooling reduction is allowed
- Computes *how much* reduction is safe

Logic Inputs

- Safety engine approval
- Overcooling detector
- Chiller power
- Cost context

Outputs

- Recommended reduction percentage
- Time window

Why this module exists

This is the **core decision engine**.

MODULE 8: Peak Demand Optimization Engine

Module Name

`enervia_peak_optimizer`

What it does

- Identifies peak demand windows
- Prioritizes optimization during peaks
- Coordinates with UPS constraints (read-only)

Inputs

- Facility power
- UPS load
- Tariff window

Outputs

- Peak mitigation recommendation

Why this module exists

This directly impacts **electricity bills**, not just kWh.

MODULE 9: Savings Estimation Engine

Module Name

`enervia_savings_engine`

What it does

- Computes:
    - kW saved
    - ₹ saved per interval
    - daily / weekly savings

Inputs

- Baseline power
- Reduced power
- Tariff
- Duration

Outputs

- Quantified savings object

Why this module exists

Optimization without numbers has **no credibility**.

MODULE 10: Recommendation Generator

Module Name

`enervia_recommendation_engine`

What it does

- Combines decisions into human-readable output
- Attaches explanations

Output Example

```
Action: Reduce cooling by 7%
Window: 18:00–20:00
Savings: 120 kW | ₹1,800
Safety: SAFE
Reason: Overcooling + Peak tariff
```

Why this module exists

Operators trust **explainable actions**, not black boxes.

MODULE 11: Audit & Logging Service

Module Name

`enervia_audit_log`

What it does

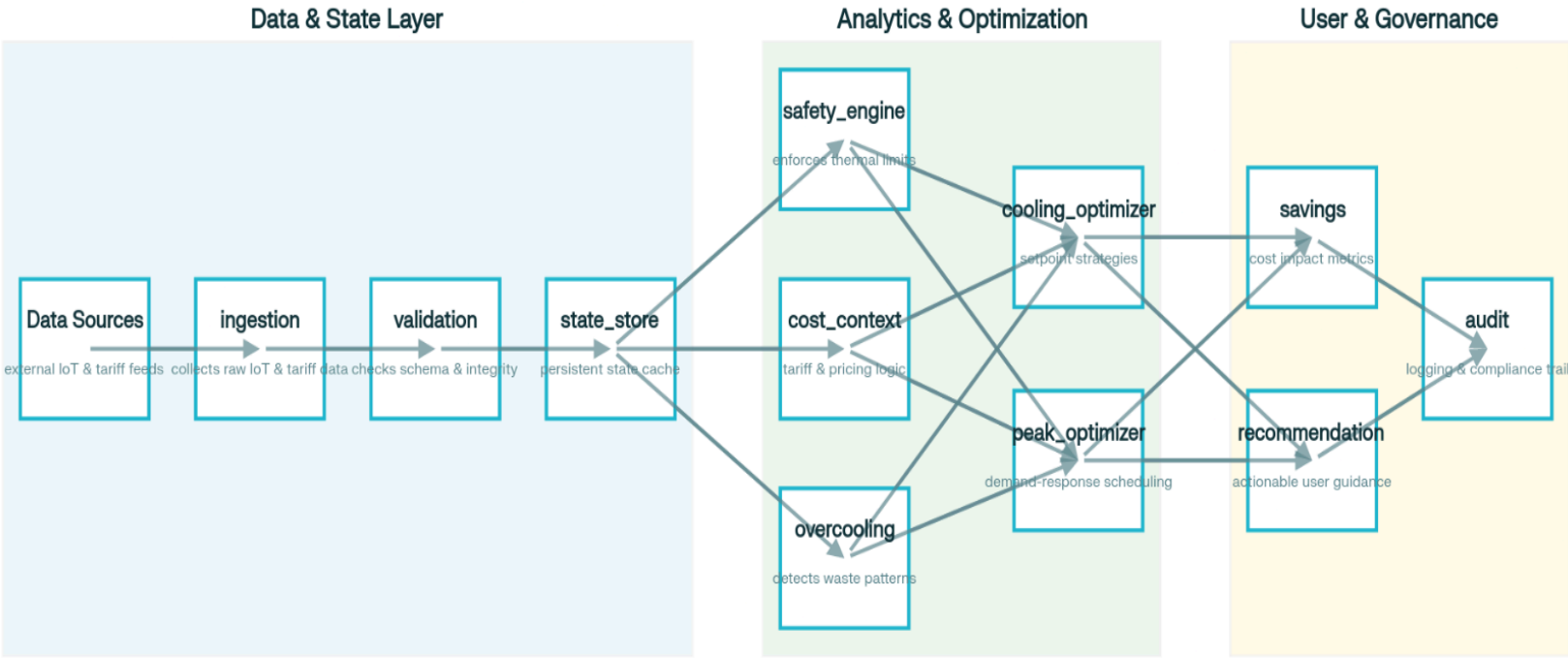
- Logs:
    - inputs
    - decisions
    - blocked actions
    - safety vetoes

Why this module exists

Critical infrastructure requires **traceability**.

Critical Design Rule

No optimization module can bypass the Safety Engine.
No recommendation is generated without savings estimation.

Enervia Backend Pipeline Architecture

Modular data flow from ingestion to governance